

Lent Week 2&3

lw703

1 Completed Task

Let p_θ be any trained model and p_d the target distribution. We aim to train a stepsize network conditioned on the input x , then apply 1-step Langevin with this step size to fine-tune samples from p_θ , *i.e.*, optimising a 1-step Langevin kernel

$$p_\gamma(y|x) = \mathcal{N}(y; x + \gamma(x)\nabla \log p_d(x), 2\gamma(x)I)$$

with $x \sim p_\theta$. To optimize this kernel, we minimize the KL divergence:

$$\gamma^* = \arg \min_{\gamma} \mathbb{D}_{\text{KL}}(p_{\gamma|\theta} \| p_d)$$

$$p_{\gamma|\theta}(y) = \int p_\gamma(y|x) p_\theta(x) dx$$

Similar to DiKL, the gradient w.r.t. γ of the KL divergence is given by:

$$\begin{aligned} \nabla_\gamma \mathbb{D}_{\text{KL}}(p_{\gamma|\theta} \| p_d) &= \int p_{\gamma|\theta}(y) (\nabla \log p_{\gamma|\theta}(y) - \nabla \log p_d(y)) \frac{\partial y}{\partial \gamma} dy \\ &= \mathbb{E}_{p_\gamma(y|x)} \mathbb{E}_{p_\theta(x)} \left[(\nabla \log p_{\gamma|\theta}(y) - \nabla \log p_d(y)) \frac{\partial y}{\partial \gamma} \right] \end{aligned}$$

where $\nabla \log p_{\gamma|\theta}(x')$ can be estimated by score matching and $\frac{\partial x'}{\partial \gamma}$ is given by autograd. Here the optimal score $\nabla_y \log p(y)$ can be analytically evaluated and used firstly, as:

$$\begin{aligned} \nabla_y \log p(y) &= \nabla_y \log \int N(y|x) p(x) dx \\ &= \frac{1}{\int N(y|x) p(x) dx} \int \nabla_y N(y|x) p(x) dx \\ &= \frac{1}{\int N(y|x) p(x) dx} \int N(y|x) p(x) \nabla_y \log N(y|x) dx \\ &= \int p(x|y) \nabla_y \log N(y|x) dx \\ &= - \int p(x|y) \frac{y - (x + \gamma \nabla \log p_d(x))}{2\gamma} dx \\ &= E\left[\frac{x + \gamma(x) \nabla \log p_d(x) - y}{2\gamma(x)} | y\right] \end{aligned}$$

$p(x|y)$ is sampled with AIS given $\nabla \log p(x|y) \propto \nabla \log p_\theta(x) + \nabla \log_\gamma(y|x)$